

Miracle Advance Technologies

HRCT

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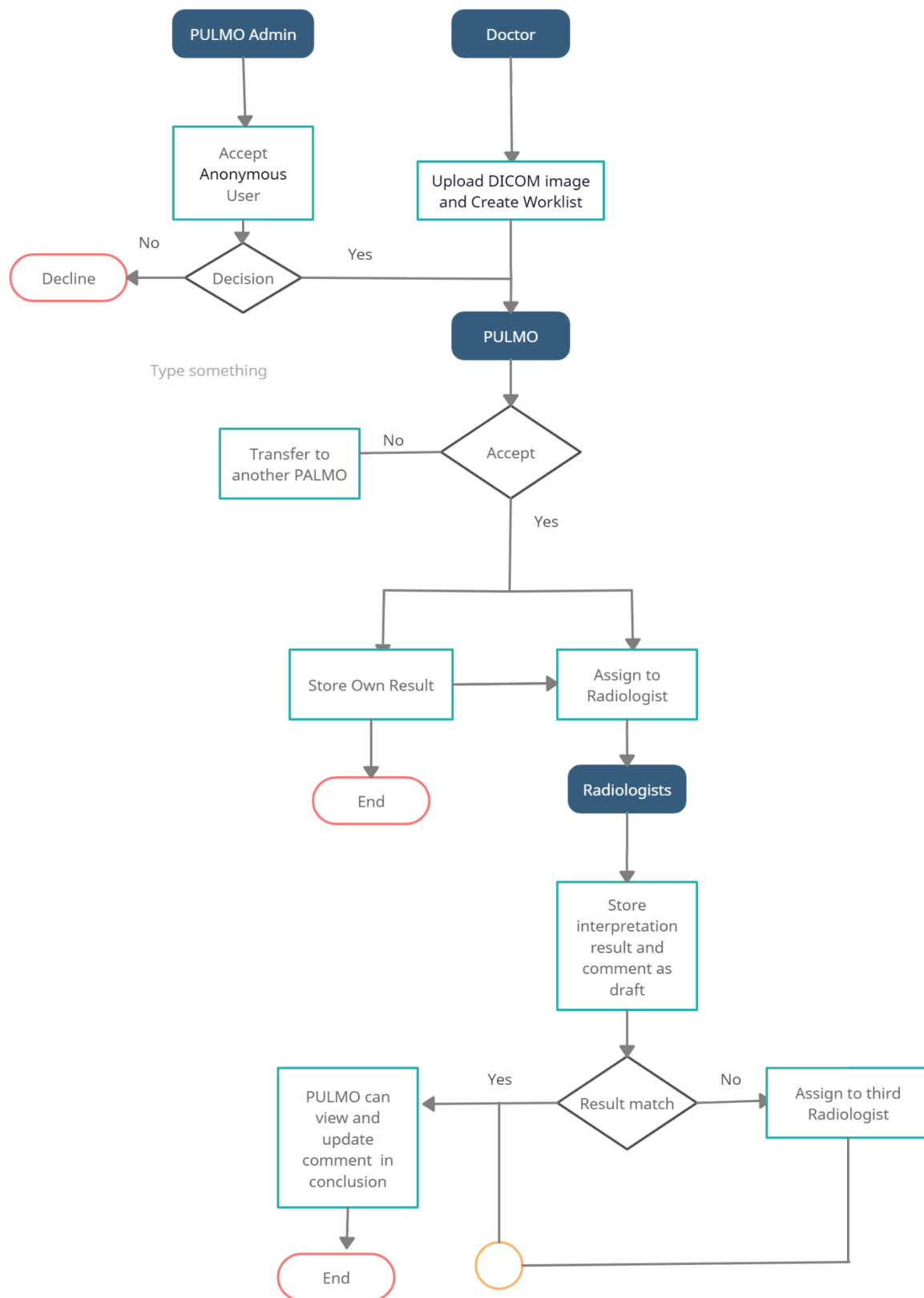
Application User Manual

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1. General Information

Easy viewing and sharing of DICOM images via HRCT-any time and any place. The solution can be set up in a matter of minutes.HRCT provides freedom to manage systems and distribute images.

UML Diagram



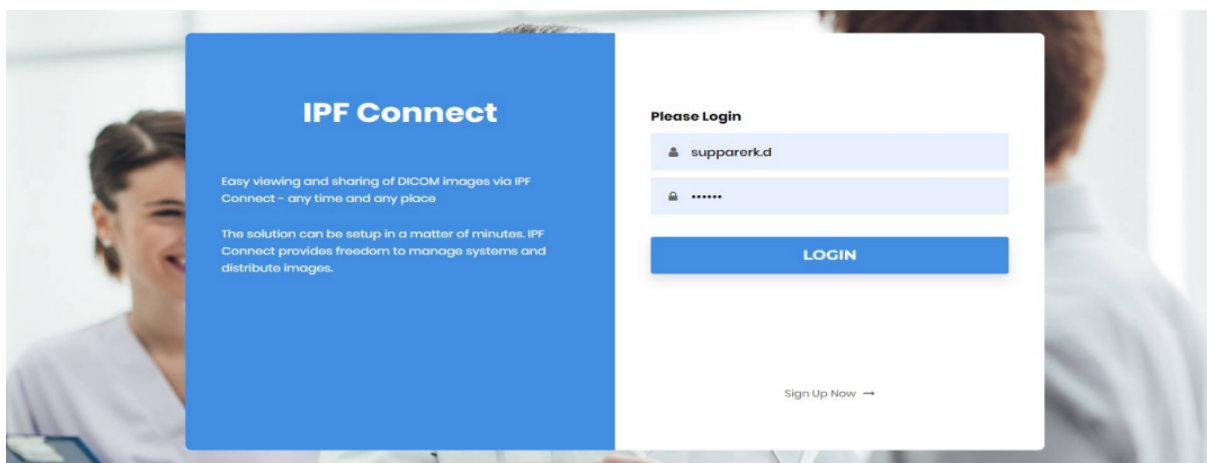
The application focuses on four users.

- **Doctors-** The doctors can upload DICOM images and create a worklist.
- **PULMO-** They can store their own results and assign radiologists.
- **RADIOLOGIST-** The radiologist can interpret results and comments as drafts.
- **PULMO Admin-** They can accept anonymous users and send the case to another PULMO.

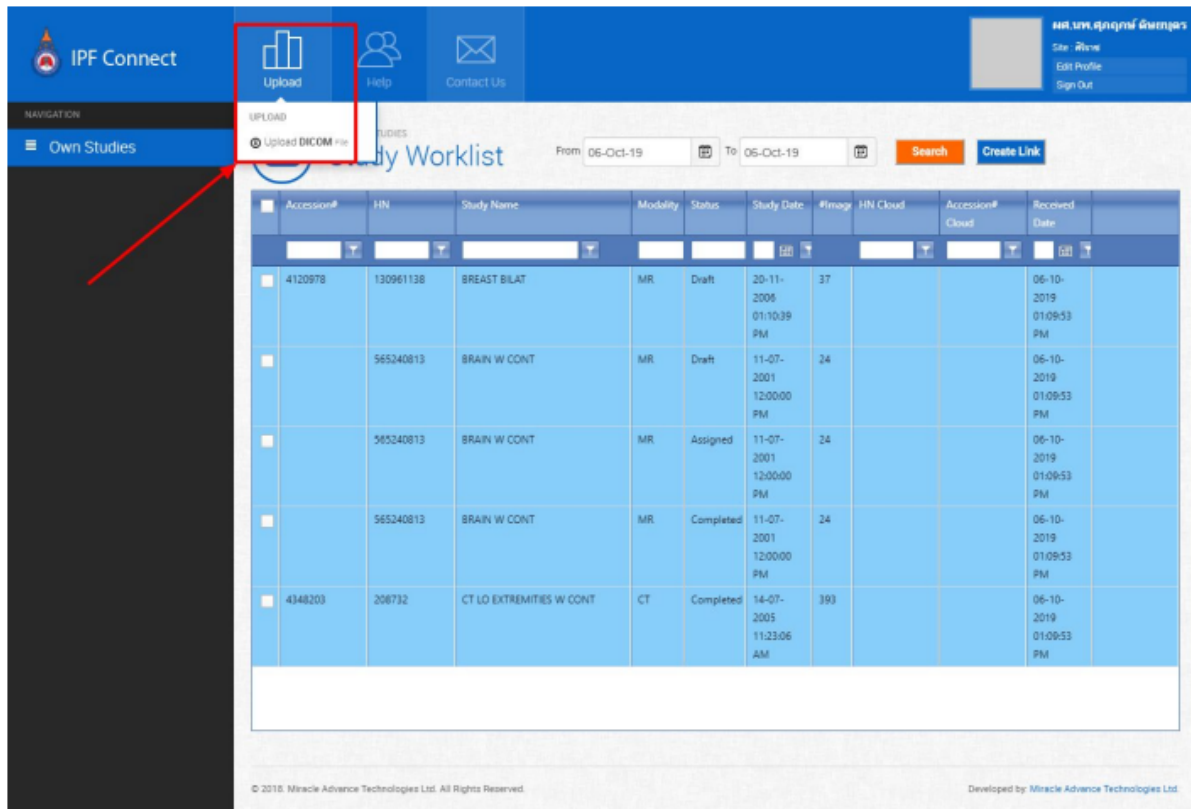
In the HRCT, the doctors login to the system and upload DICOM images to create a worklist. These worklists will be sent to any PULMO (randomly) to finalise the result. The doctors will only see the PULMO's result and conclusion comment. The PULMO can assign this study between the first and second radiologist. The two radiologists are able to see their assigned study. After successfully assigned the radiologists, PULMO is able to see assigned radiologists in the conclusion tab. Once the first radiologist and second radiologist will finalize their result and if it is not the same, a third radiologist will be assigned. Once the first, second and third radiologist will finalize their results, PULMO will be able to see their results and will be able to update PULMO's result and enter a conclusion comment and will finalize. After finalizing the result by PULMO, the doctor is able to see the final result. The Anonymous user can create a study worklist, but it will auto disappear if the PULMO admin denies the case.

Login as hospital Source

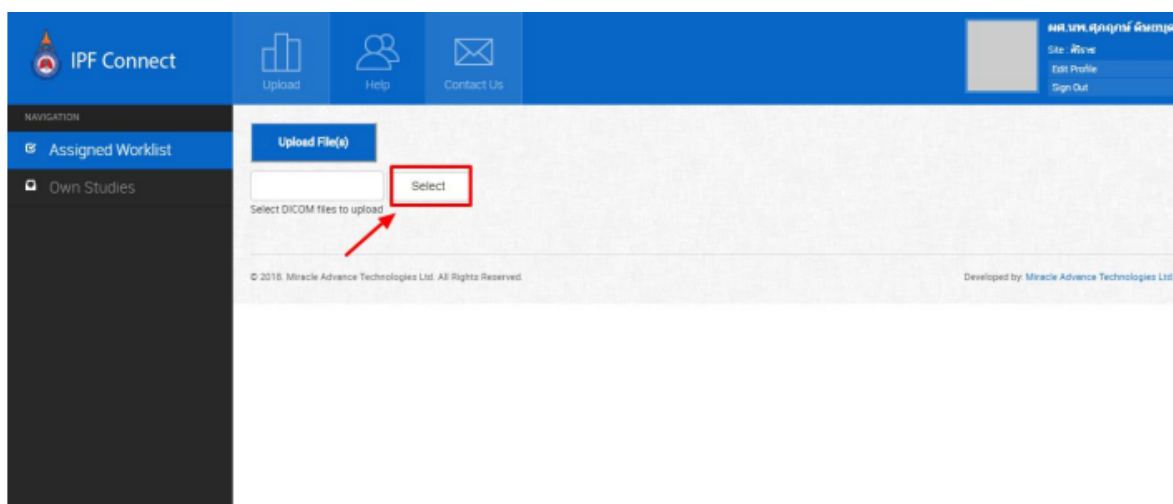
Hospital source user will login using his/her credentials (user id and password).



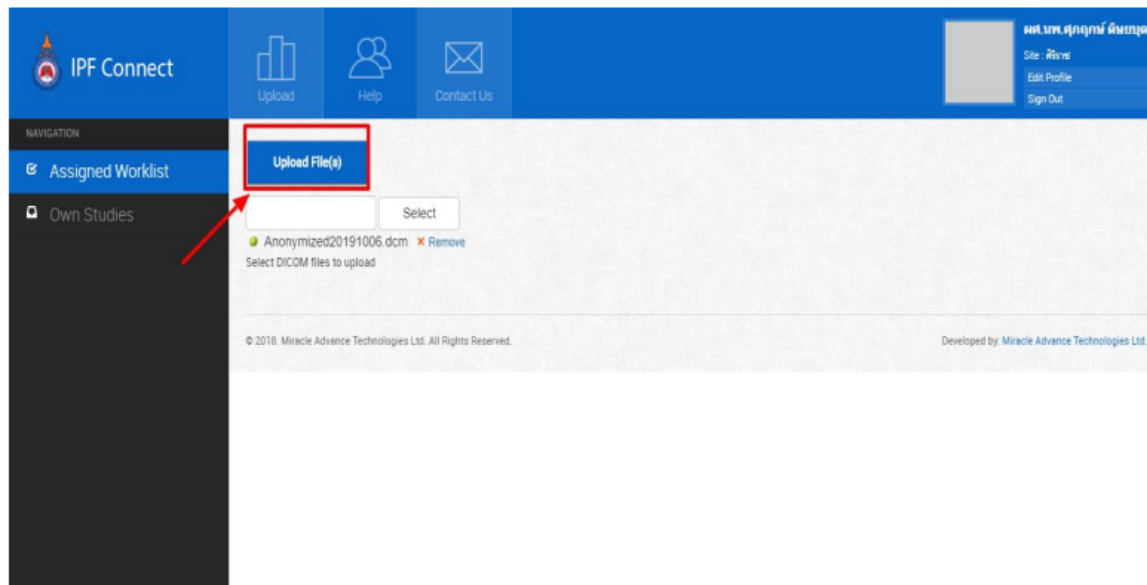
After successful login, the source user will get his/her dashboard. Here users will be able to upload DICOM images to create a worklist.



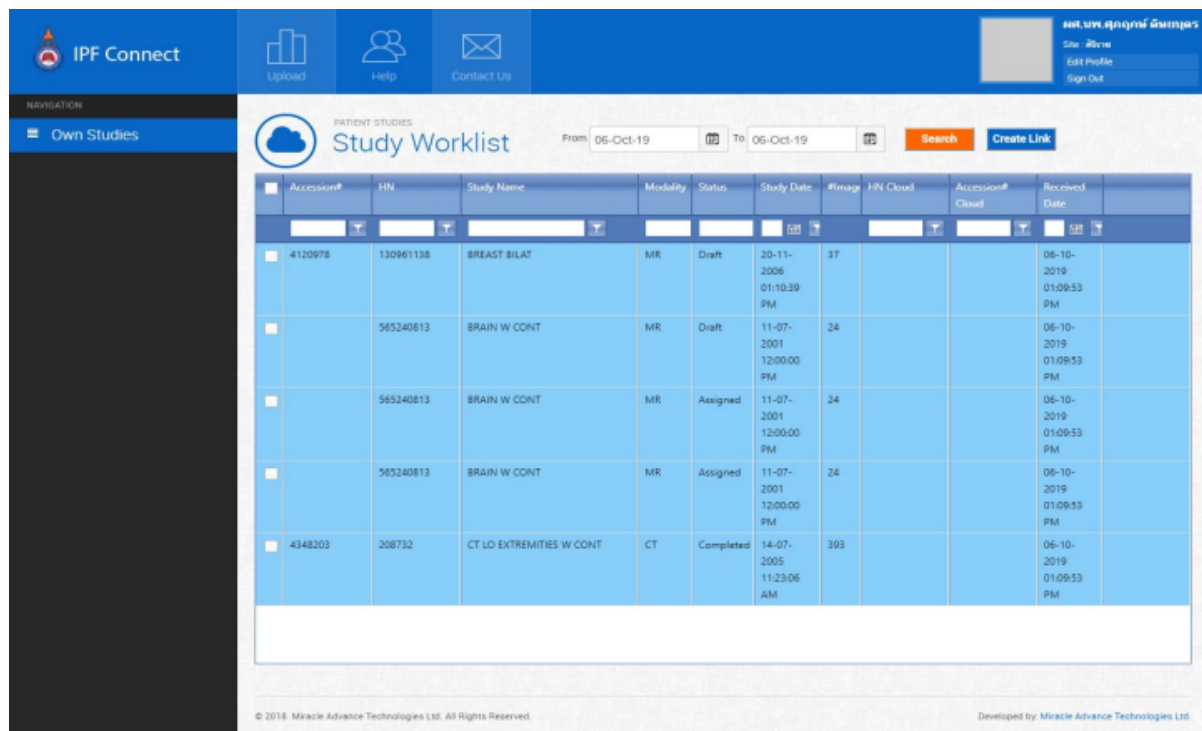
After clicking the **Upload DICOM File** button, the system will ask to browse the dicom image. Users need to click the **“Select”** button to browse DICOM image from his/her device.



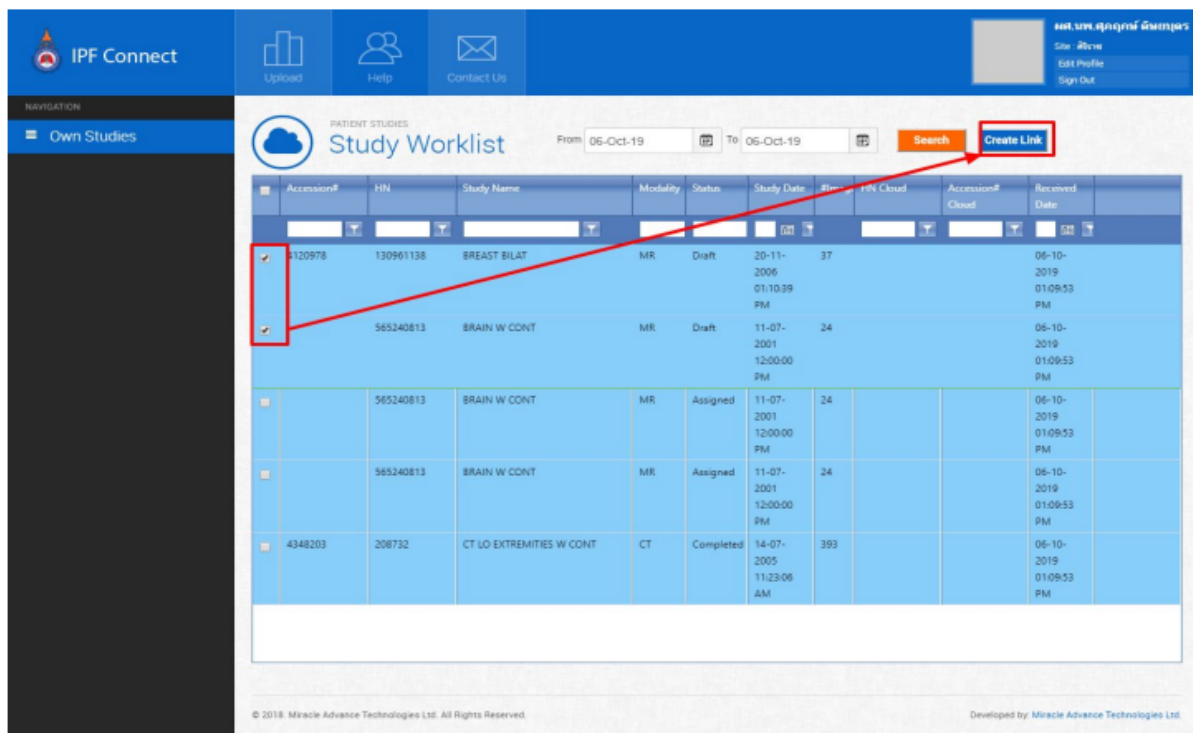
Once a user will select a DICOM image from his/her device, need to click **Upload Files** button to upload image and a study worklist will created.



Then the user can able to see his/her created worklist.



Now users can able to create a link between the two studies.



Users are able to store assessment data as a draft or finalise by double clicking the worklist. Once the user will double click on the worklist then the system will open a dialog box to store **assessment data**. For finalisation assessment data, he/she must have to click at least one radio button.

Upload

Help

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Site: ไซต์

Edit Profile

Sign Out

NAVIGATION

Own Studies

PATIENT STUDIES

Study Worklist

From 06-Oct-19

To 06-Oct-19

Search

Create Link

Accession#	HN	Study Name	Modality	Status	Study Date	#Image	HN Cloud	Accession# Cloud	Received Date
4120978	130961138	BREAST BILAT	MR	Draft	20-11-2006 01:10:39 PM	7			06-10-2019 01:09:53 PM
	565240813	BRAIN W CONT	MR	Draft	11-07-2001 12:00:00 PM	24			06-10-2019 01:09:53 PM
	565240813	BRAIN W CONT	MR	Assigned	11-07-2001 12:00:00 PM	24			06-10-2019 01:09:53 PM
	565240813	BRAIN W CONT	MR	Assigned	11-07-2001 12:00:00 PM	24			06-10-2019 01:09:53 PM
4348203	208732	CT LO EXTREMITIES W CONT	CT	Completed	14-07-2005 11:23:06 AM	393			06-10-2019 01:09:53 PM

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Double Click

Upload

Help

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Edit Profile

Sign Out

NAVIGATION

Own Studies

Close

Save as Draft

Submit Assessment Data

Referral

ลักษณะทางคลินิก

Pulmonary function tests(PFT)

Serology

Referral-required document

การวินิจฉัย (ระบุ)

การรักษา

อื่นๆ (ระบุ)

โปรดระบุ

Referral checklist

ใช่	ไม่ใช่	Clinical Presentation
☒	☐	ผู้ป่วยที่อายุมากกว่า 40 ปีขึ้นไป
☒	☐	ผู้ป่วยมีอาการหอบเหนื่อยมากขึ้น (progressive dyspnea) หรือเหนื่อยเวลาออกแรง (Exertional Dyspnea) มากกว่า 2 เดือนขึ้นไป โดยที่หาสาเหตุไม่พบ
☐	☐	ผู้ป่วยมีอาการไอเรื้อรัง แบบไม่มีเสมหะ มากกว่า 2 เดือน
☐	☐	Chest Radiography มีลักษณะผิดปกติแบบ reticular infiltration ซึ่งเด่นที่ชายปอดส่วนล่างทั้ง 2 ข้าง
ใช่	ไม่ใช่	ประวัติและอาการ
☒	☐	อาการของ connective tissue disease เช่น ข้ออักเสบ, ท้องแข็ง, ผู้ป่วยอัมพาตไม่ขึ้น, กล้ามเนื้ออ่อนแรง, ลาแห่ง, ปากแห้ง และ Raynaud's phenomenon
		โปรดระบุ

Assessment.aspx?studyID=376&worklist=study&isConf=False&comments=Text&headRad=

ologies Ltd.

IPF Connect

Assessment Form

Close Save for Later Submit

Clinical Data 1 Clinical Data 2 Pulmonary Function Tests Serology

ใช่ ไม่ใช่ ประวัติ

ประวัติการได้รับยาทางชนิด เช่น

- ☐ ☐ - Amiodarone
โปรดระบุ
- ☐ ☐ - Bleomycin
โปรดระบุ
- ☐ ☐ - Nitrofurantoin
โปรดระบุ
- ☐ ☐ - Methotrexate
โปรดระบุ
- ☐ ☐ - Gold salts
โปรดระบุ
- ☐ ☐ - ยาอื่นที่สงสัย
โปรดระบุ

ใช่ ไม่ใช่ ตรวจร่างกาย

Assessment.aspx?studyID=14632&workliststudy&isConf=false&comments= &headRad= &

Received Date
05 09/12/2020
12:44:16 PM

พญ.สิริยุภา อภิชาติ
Date: Saraburi
Edit Profile
Sign Out

Referral ประวัติการตรวจ Pulmonary function tests(PFT) Serology Interpretation

Laboratory Parameters	Result	Unit	Not Performed
FVC (pre-BD)		% predicted	☐
FEV1 (pre-BD)		% predicted	☐
TLC		% predicted	☐
DLCO		% predicted	☐
DLCO/VA		% predicted	☐
6MWD		Metres	☐
SpO2 (pre-6MWT)		%	☐
SpO2 (post-6MWT)		%	☐

Note: The user must fill up every section of Clinical Data1 and Clinical Data 2.

The user can fill up the data later and press the button **Save for later**. After finishing the assessment form the user has to press the **Submit** button.

Once PULMO will finalise the result, source user will only see the head node's result and conclusion comment.

NAVIGATION

Own Studies

Close

Node Head:

Radiologist Name : พญ. ศิริยุวศา

Site Name : ศิริราช

Comment : conclusion comments by rad head

Result

Result	UIP	Probable UIP	Early UIP	Truly indeterminate	Alternative diagnosis
Rad head result	☒	☐	☐	☐	☐

Conclusion Text : conclusion comments by rad head

RadiologyInfo.aspx?studyID=367&confirm=True&comments=conclusion comments by rad head&headRadu2

Accession#	Received Date	Rad Info
06-10-2019 01:09:53 PM		Rad Info
06-10-2019 01:09:53 PM		Rad Info
06-10-2019 01:09:53 PM		Rad Info
06-10-2019 01:09:53 PM		Rad Info

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Login as PULMO

PULMO user will login using his/her credentials (user id and password).

IPF Connect

Easy viewing and sharing of DICOM images via IPF Connect - any time and any place

The solution can be setup in a matter of minutes. IPF Connect provides freedom to manage systems and distribute images.

Please Login

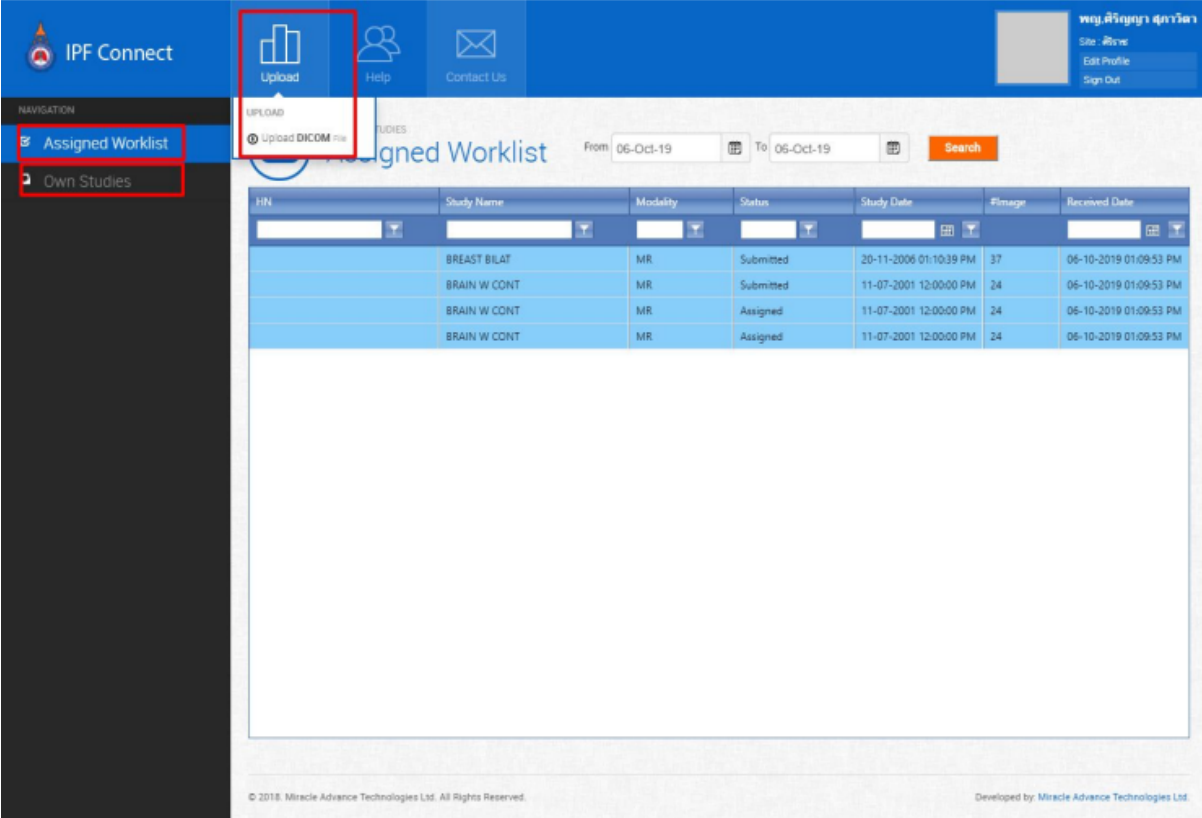
sirinya.s

.....

LOGIN

Sign Up Now →

After successful login, PULMO users will get his/her dashboard. Here the user will be able to upload dicom image to create worklist and he/she will also get worklist which has been already assigned to him/her.



IPF Connect

NAVIGATION

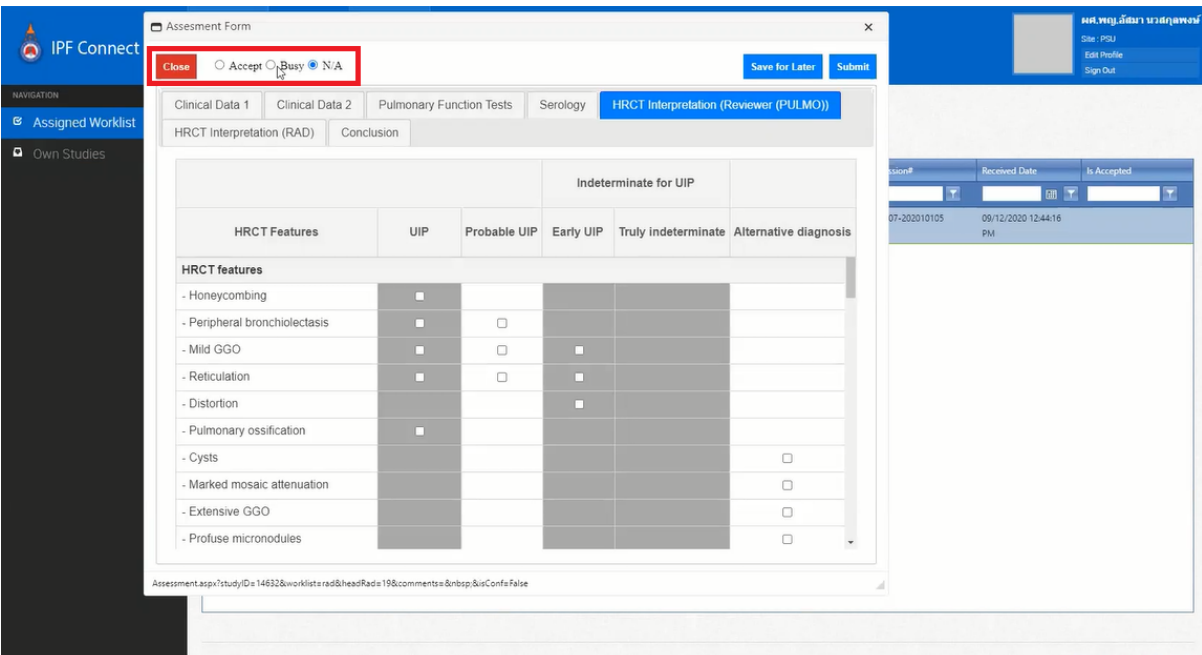
- Assigned Worklist
- Own Studies

Assigned Worklist

From: 06-Oct-19 To: 06-Oct-19 Search

HN	Study Name	Modality	Status	Study Date	#Image	Received Date
	BREAST BILAT	MR	Submitted	20-11-2006 01:10:39 PM	37	06-10-2019 01:09:53 PM
	BRAIN W CONT	MR	Submitted	11-07-2001 12:00:00 PM	24	06-10-2019 01:09:53 PM
	BRAIN W CONT	MR	Assigned	11-07-2001 12:00:00 PM	24	06-10-2019 01:09:53 PM
	BRAIN W CONT	MR	Assigned	11-07-2001 12:00:00 PM	24	06-10-2019 01:09:53 PM

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Assessment Form

Close Accept **Busy** N/A Save for Later Submit

Clinical Data 1 Clinical Data 2 Pulmonary Function Tests Serology **HRCT Interpretation (Reviewer: PULMO)**

HRCT Interpretation (RAD) Conclusion

HRCT Features	UIP	Probable UIP	Indeterminate for UIP		Alternative diagnosis
			Early UIP	Truly indeterminate	
- Honeycombing	☒				
- Peripheral bronchiolectasis	☒	☐			
- Mild GGO	☒	☐	☒		
- Reticulation	☒	☐	☒		
- Distortion			☒		
- Pulmonary ossification	☒				
- Cysts					☐
- Marked mosaic attenuation					☐
- Extensive GGO					☐
- Profuse micronodules					☐

Assessment.aspx?studyID=14632&worklist=rad&headRad=19&comments=&isConf=0

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The Pulmo can press Accept, Busy, and N/A. He/She has to click the button sincerely. If they select N/A or Busy, the study will transfer to another Palmo.

If the user will double click in assigned worklist, he/she can only store own result or can assigned this study between first and second radiologist.

Close

Save as Draft Finalize Interpretation

Referral Interpretation Conclusion

* Self Reporting Assign Radiologist

HTCT Features	Indeterminate for UIP				Alternative diagnosis
	UIP	Probable UIP	Early UIP	Truly indeterminate	
HRCT features					
- Honeycombing	☐				
- Peripheral bronchiolectasis	☐	☐			
- Mild GGO	☐	☐	☒		
- Reticulation	☐	☐	☐		
- Distortion			☐		
- Pulmonary ossification	☐				
- Cysts					☐
- Marked mosaic attenuation					☐
- Extensive GGO					☐

Assessment.aspx?studyID=367&worklist=rad&headRad=2&comments=Test&isConf=False

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Once user will finalise the interpretation result, he/she can able to see result in conclusion tab.

Close

Referral Interpretation Conclusion

Node Head:

Radiologist Name : หมอ. ศิริกฤตา
Site Name : ศิริราช
Comment : Comments by rad head

Result

Result	UIP	Probable UIP	Early UIP	Truly indeterminate	Alternative diagnosis
Node head result	☐	☐	☒	☐	☐

Conclusion Text :

Assessment.aspx?studyID=347&worklist=rad&headRad=2&comments=My testing is done&isConf=False

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Once a user will assign the study between two radiologists, two radiologists can able to see their assigned study.

IPF Connect

NAVIGATION

- Assigned Worklist
- Own Studies

Close

Save as Draft

Assign Radiologist

Referral

ลักษณะทางคลินิก

Pulmonary function tests(PFT)

Serology

Interpretation

Conclusion

Self Reporting

Assign Radiologist

HTCT Features	UIP	Probable UIP	Indeterminate for UIP		
			Early UIP	Truly Indeterminate	Alternative diagnosis
HRCT features					
- Honeycombing	☒				
- Peripheral bronchiolectasis	☐	☐			
- Mild GGO	☐	☒	☐		
- Reticulation	☒	☐	☐		
- Distortion			☒		
- Pulmonary ossification	☐				
- Cysts					☐

After successful assigned the radiologists, PULMO can able to see assigned radiologists in conclusion tab.

IPF Connect

NAVIGATION

- Assigned Worklist
- Own Studies

Close

Save as Draft

Assign Radiologist

Referral

ลักษณะทางคลินิก

Pulmonary function tests(PFT)

Serology

Interpretation

Conclusion

Node Head:

Radiologist Name : พญ. ศิริยุพดา

Site Name : ศิริราช

Comment : commented by node head

Rad 1:

Radiologist Name : รศ. นพ. ชีรกร

Site Name : สงฆ์ถนอมสินทรัพย์

Comment :

Rad 2:

Once first radiologist and second radiologist will finalise their result and if it is not the same, third radiologist will be assigned.

Close

Referral ลักษณะทางคลินิก Pulmonary function tests(PFT) Serology Interpretation **Conclusion**

Radiologist Name : รศ. นพ. ชัยกร
 Site Name : สงขลาเคหะราษฎร์
 Comment : rad comments 2

Rad 3:

Radiologist Name : รศ. นพ. ชัยกร
 Site Name : สงขลาเคหะราษฎร์
 Comment :

Result

Result	UIP	Probable UIP	Early UIP	Truly indeterminate	Alternative diagnosis
Node head result	☒	☐	☐	☐	☐

AssessmentForHead.aspx?studyID=304&worklist=rad&headRad=2&comments=Test&isConf=False

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Once the first, second and third radiologist will finalise their results, PULMO users will be able to see their results and be able to update PULMO's result and enter conclusion comments and will finalise. After finalise result by PULMO, source can able to see the final result.

Close

Referral ลักษณะทางคลินิก Pulmonary function tests(PFT) Serology Interpretation **Conclusion**

Result

Result	UIP	Probable UIP	Early UIP	Truly indeterminate	Alternative diagnosis
Node head result	☒	☐	☐	☐	☐
Rad 1 result	☐	☐	☒	☐	☐
Rad 2 result	☐	☐	☐	☐	☐
Rad 3 result	☐	☐	☐	☒	☐

Conclusion text : Conclusion text by head of node

Save as Final

AssessmentForHead.aspx?studyID=304&worklist=rad&headRad=2&comments=Test&isConf=False

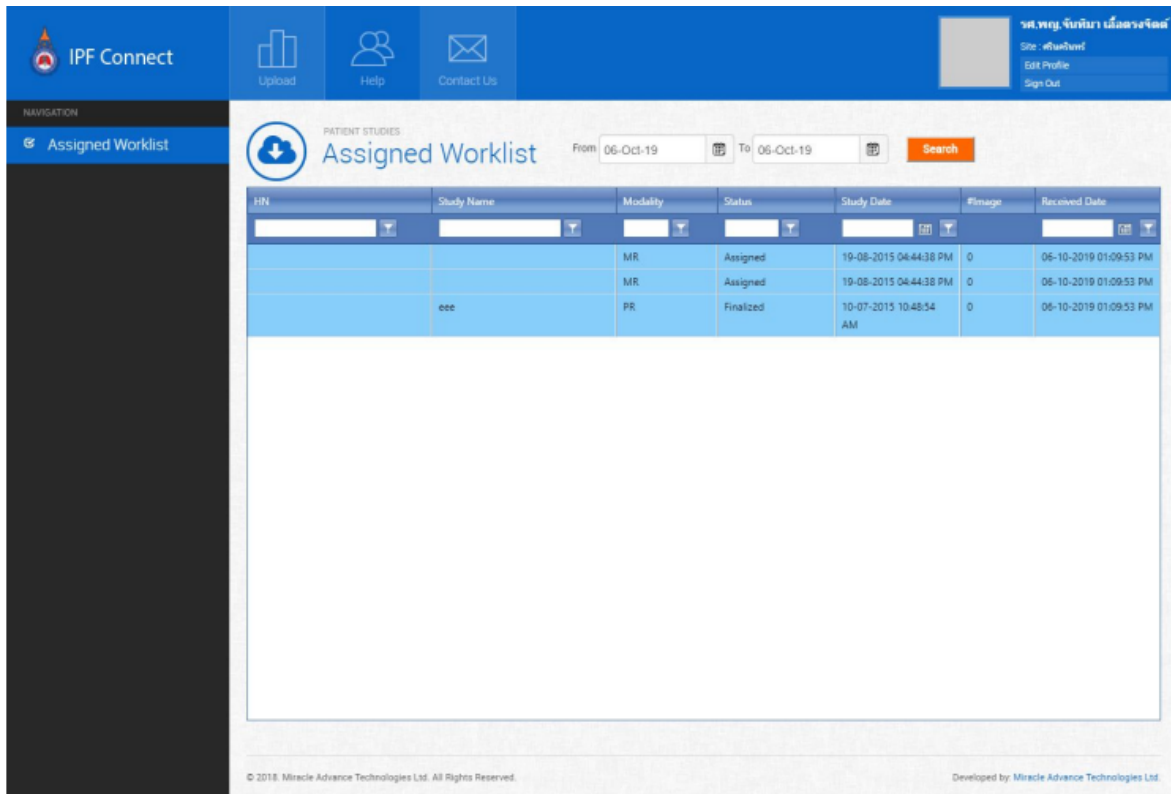
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Login as Radiologist

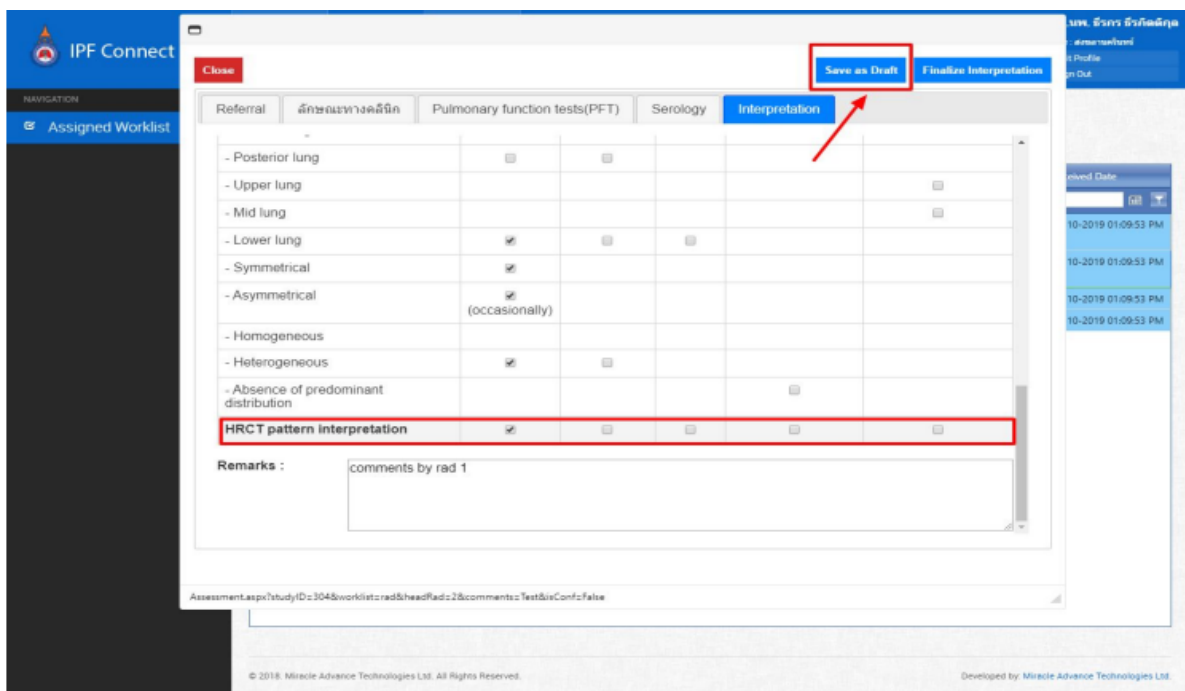
Radiologist user will login using his/her credentials (user id and password).



After successful login, the radiologist user will get his/her dashboard. Here user/Pulmo will be able to see the studies which has been assigned to him/her.



After double clicking on study worklist, the radiologist can able to store interpretation result and comment as draft.



Once user will finalise the study result, PULMO can able to see the result and comments in their worklist.

IPF Connect

Close Save as Draft Finalize Interpretation

Referral ลักษณะทางคลินิก Pulmonary function tests (PFT) Serology Interpretation

- Posterior lung	☐	☐		
- Upper lung				☐
- Mid lung				☐
- Lower lung	☒	☐	☐	
- Symmetrical	☒			
- Asymmetrical	☒	(occasionally)		
- Homogeneous				
- Heterogeneous	☒	☐		
- Absence of predominant distribution				☐
HRCT pattern interpretation	☒	☐	☐	☐

Remarks : finalize comments by rad 1

Assessment.aspx?studyID=304&worklist=rad&headRad=2&comments=Test&isConf=false

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NOTE: The Anonymous user can create a study worklist, but it will auto disappear if the PULMO admin denies the case.

NOTE: The doctor can create a study worklist, but it will auto transfer to another PULMO when PULMO delays accepting that case for seven days.